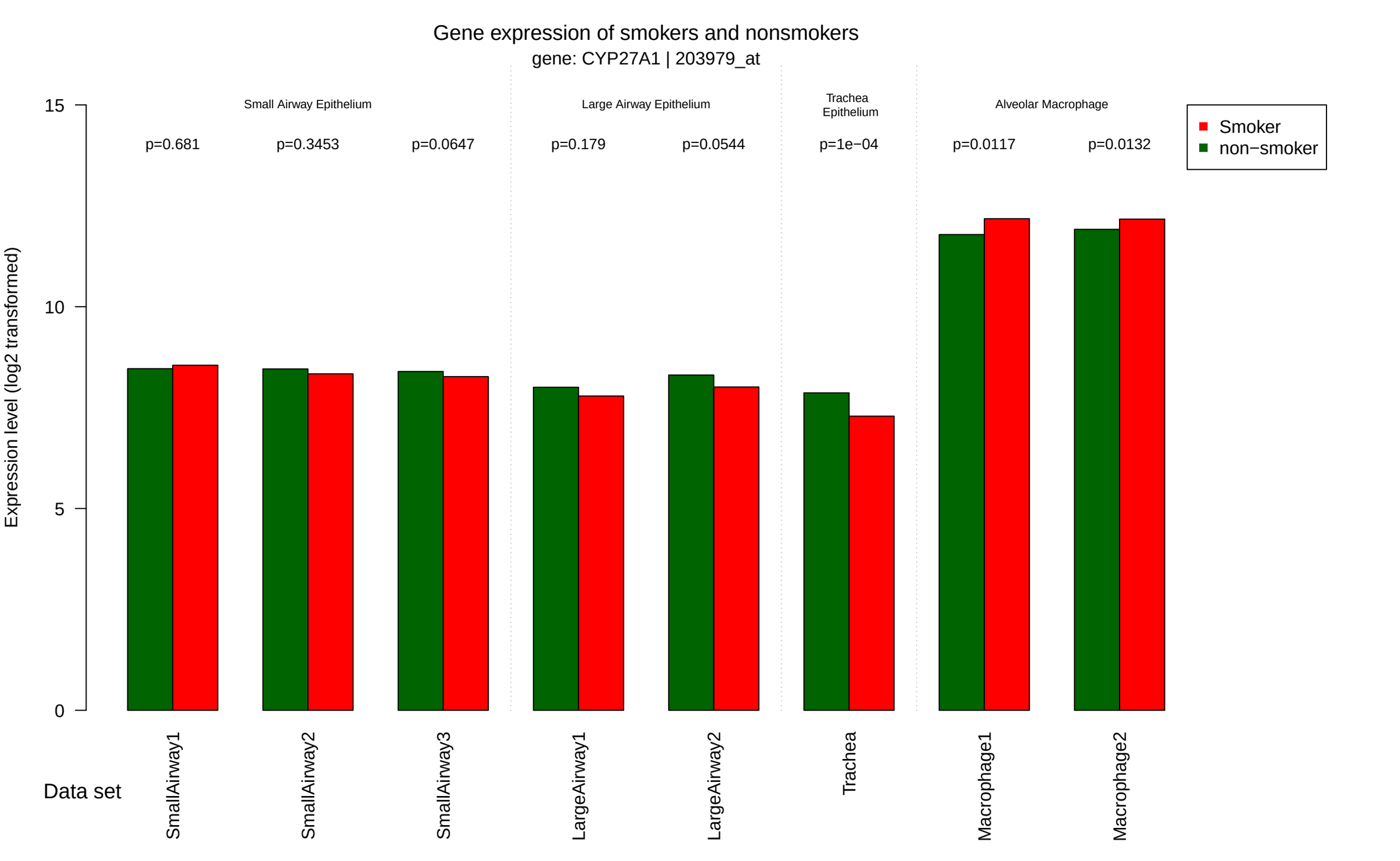
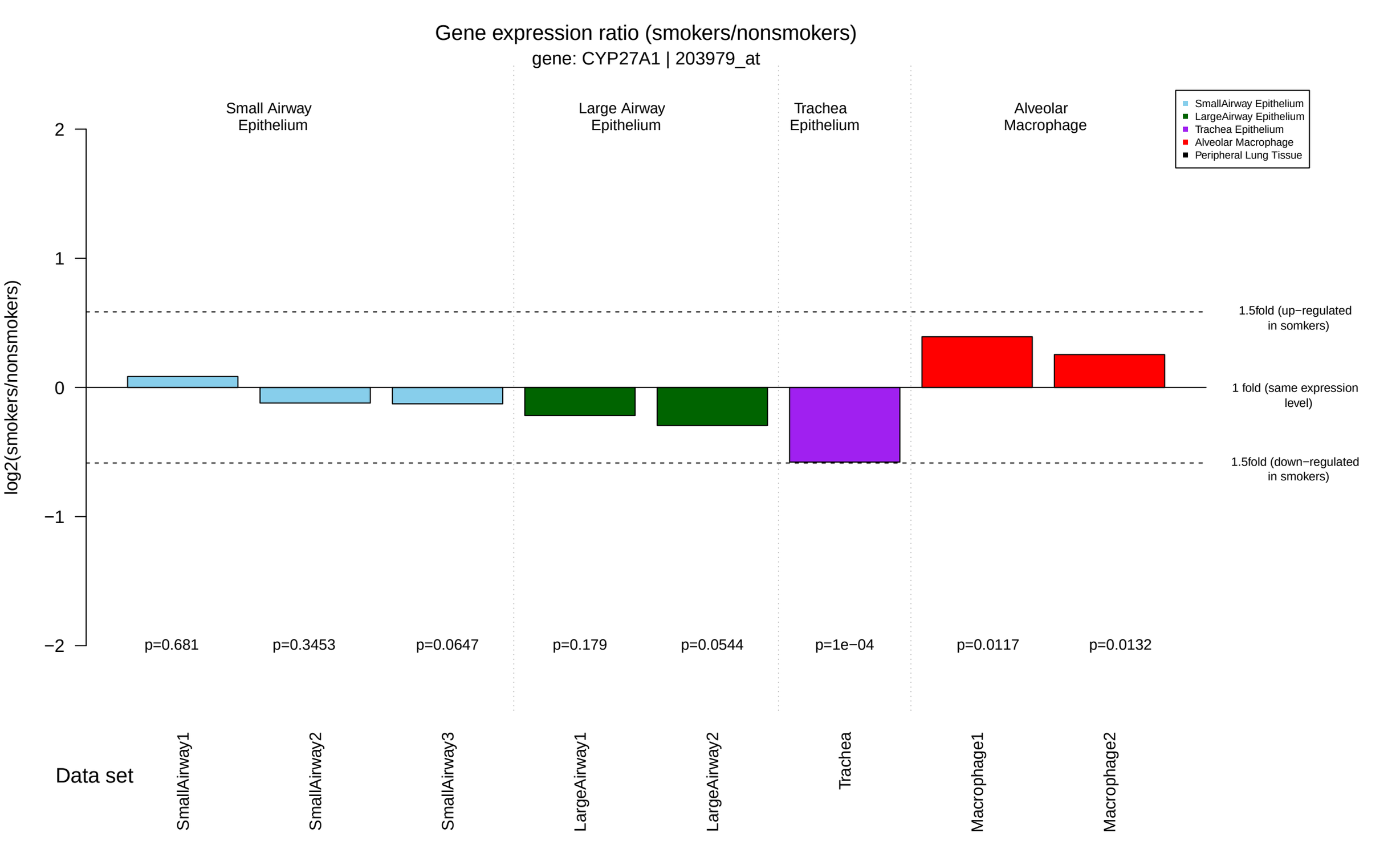
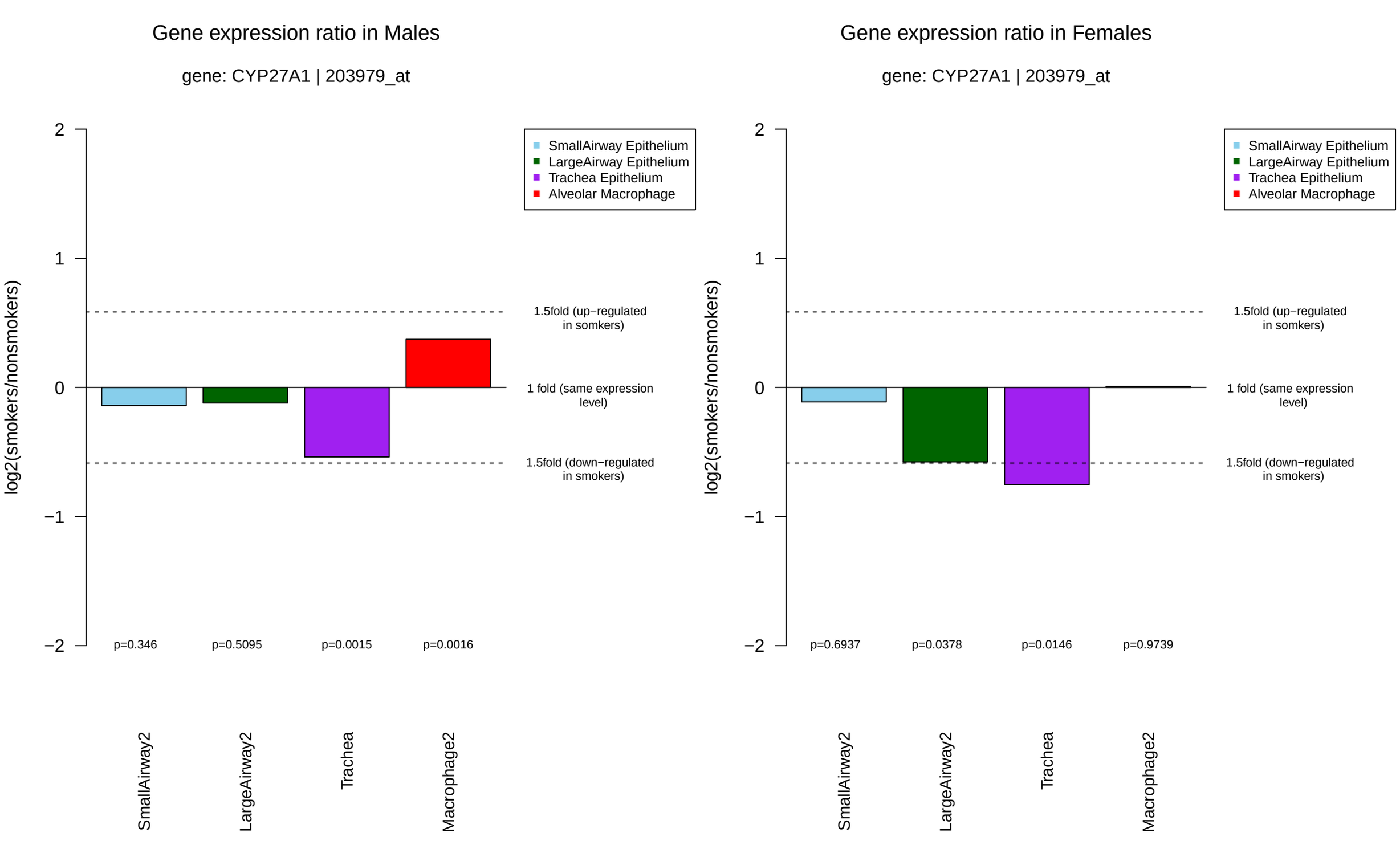
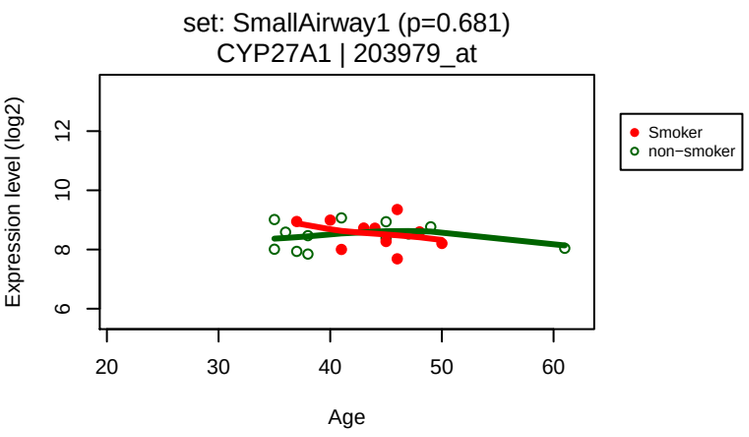
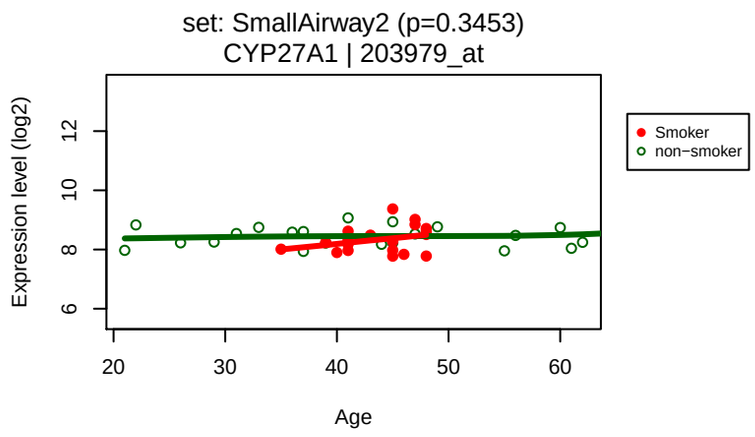
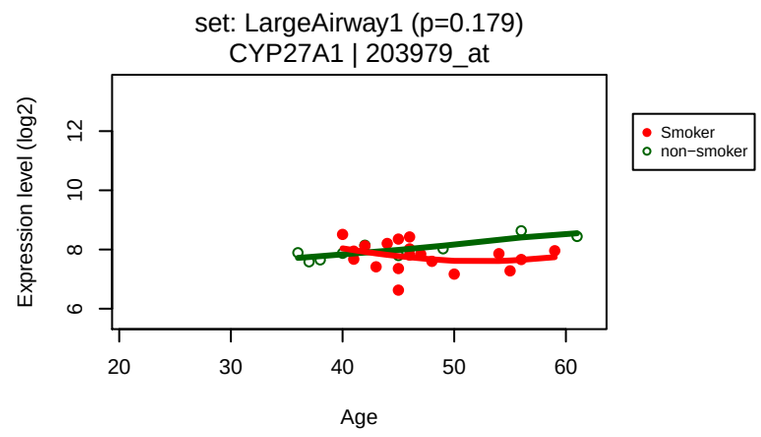

Gene expression of smokers and nonsmokers
gene: CYP27A1 | 203979_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.681
p=0.3453
p=0.0647
p=0.179
p=0.0544
p=1e−04
p=0.0117
p=0.0132
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

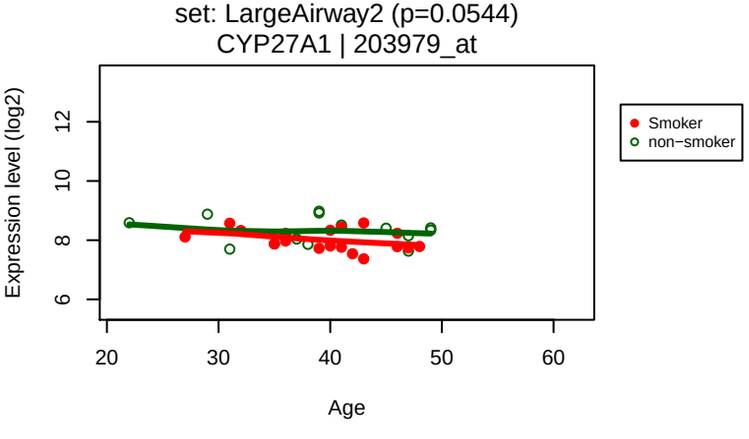

Gene expression ratio (smokers/nonsmokers)
gene: CYP27A1 | 203979_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.681
p=0.3453
p=0.0647
p=0.179
p=0.0544
p=1e−04
p=0.0117
p=0.0132
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

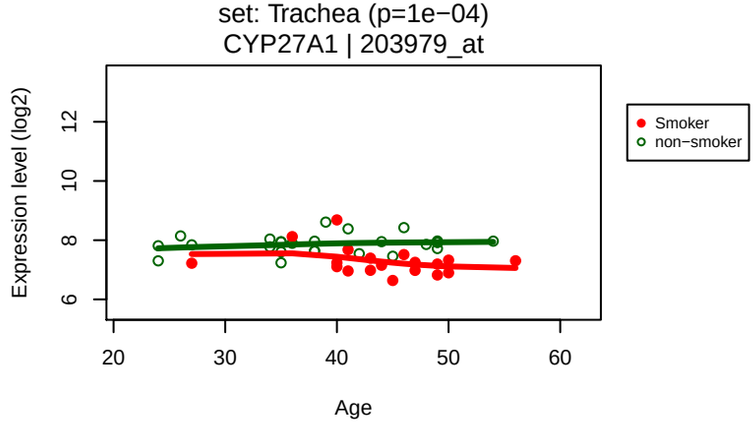

Gene expression ratio in Males
gene: CYP27A1 | 203979_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.346
p=0.5095
p=0.0015
p=0.0016
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP27A1 | 203979_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6937
p=0.0378
p=0.0146
p=0.9739
SmallAirway2
LargeAirway2
Trachea
Macrophage2

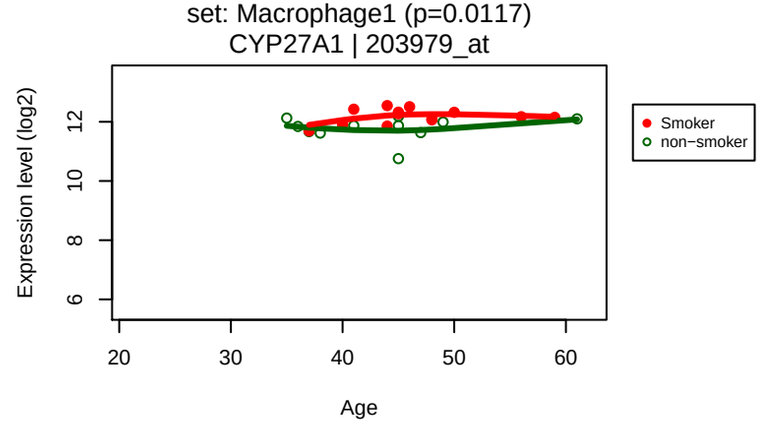

set: SmallAirway1 (p=0.681)
CYP27A1 | 203979_at
Expression level (log2)
Age
Smoker
non−smoker

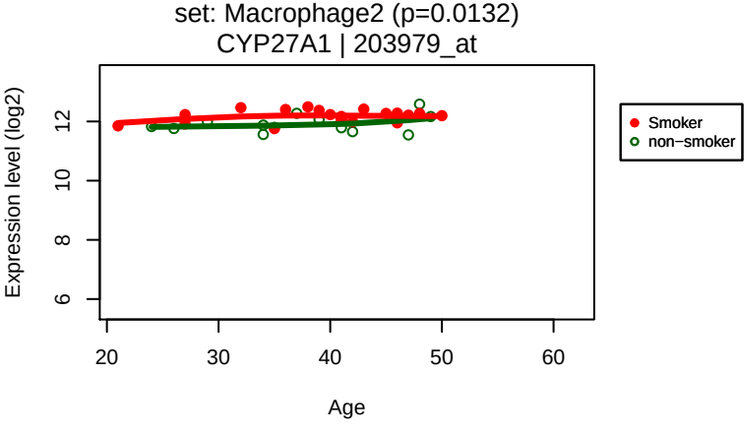

set: SmallAirway2 (p=0.3453)
CYP27A1 | 203979_at
Expression level (log2)
Age
Smoker
non−smoker

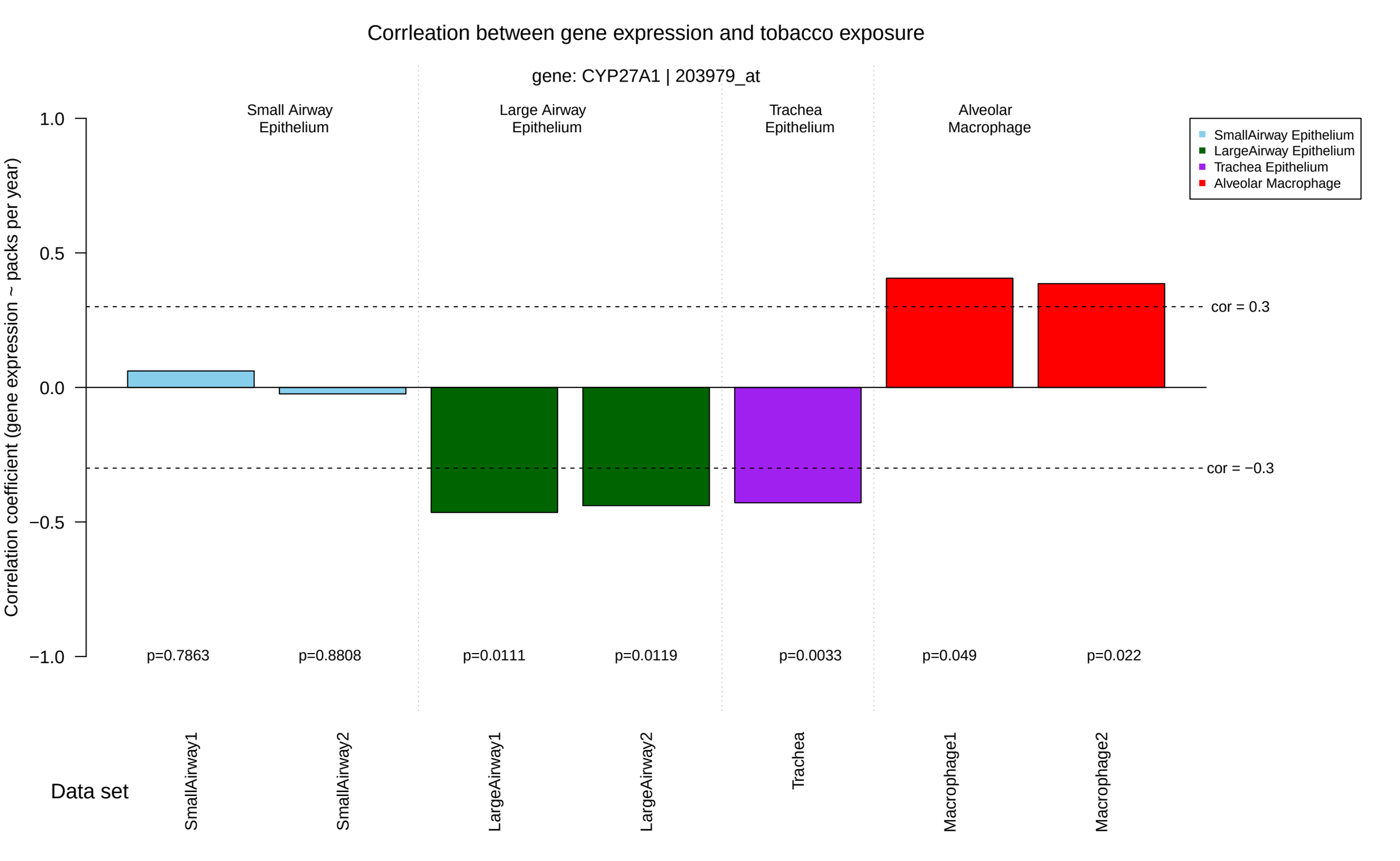

set: LargeAirway1 (p=0.179)
CYP27A1 | 203979_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0544)
CYP27A1 | 203979_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
CYP27A1 | 203979_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0117)
CYP27A1 | 203979_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0132)
CYP27A1 | 203979_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP27A1 | 203979_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7863
p=0.8808
p=0.0111
p=0.0119
p=0.0033
p=0.049
p=0.022
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2